

Ali Nawaf

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Education

Case Western Reserve University | Cleveland, OH

- Bachelors of Science in Computer Science, Secondary Major: Mathematics. May 2027
- Masters of Science in Machine Learning and Artificial Intelligence. May 2027

Course work: HPC, Linear Algebra, Security, Networks, Data structures, Algorithms, Probability

Experience

Machine Learning Research Assistant | Huston Methodist Hospital | remote Aug 2023 - Present

- Researched and developed a novel hypothesis for cardiovascular risk prediction, applying a ResNet CNN to test the theory against a real-world ECG scan dataset.
- Engineered a signal processing pipeline to digitize raw ECG signals, implementing open-source models and post-processing algorithms to extract meaningful signals and enhance fidelity for downstream ML analysis.

Software Engineer | Eaton | Cleveland, OH Jan 2025 – May 2025

- Engineered an automated Azure data pipeline using SQL queries and Python scripts to process, clean, and aggregate over 100,000 Salesforce records monthly, cutting data processing latency by 35%.
- Led the team to present project results, metrics and documentation to supervisors and non-technical leadership, facilitating data-driven decision-making.

Machine Learning Intern | Heads-up Hockey | Waterford, VA Dec 2024 – Aug 2025

- Developed an interactive Swift-based iOS application, integrating a Core-ML model for efficient on-device inference which improved user retention by 63%.
- Optimized the Core-ML model for on-device performance by applying 16-bit float quantization, reducing model size by 50% and improving battery-life efficiency.
- Leveraged the Core-ML framework to accelerate the model on Apple's SOC hardware (Neural Engine), achieving real-time inference.
- Automated a computer vision data annotation pipeline using Kubernetes, reducing manual data preparation time by 70% and accelerating model training cycles.

Projects

Medical Smart Labeler | PyTorch, Python, API, git, GitHub, Docker, Networking | [link](#)

- Reduced manual image annotation time by 60% by developing a Python-based tool that integrated with the Label-Studio API to pre-label images using a fine-tuned YOLOv10 model.
- Analyzed and pruned redundant layers from the YOLOv10 model, reducing computational complexity and inference latency by 30% for a real-time pre-labeling tool.

Electro Vector App | Swift, Python, git, GitHub, API

- Empowered medical staff with faster diagnostic insights by developing an iOS app that transforms raw ECG signals into clinical Vectorcardiograms, automatically extracting 5+ key cardiac risk metrics.

Network Router Simulator | C, Binary I/O, Pcap Parsing |

- Developed a flow analyzer (hash-based NetFlow, TCP RTT), a router simulator, and an IPv4 binary trace parser using C to quantitatively analyze complex network packet data.

Skills

Languages: Python, Swift, C++, C, CUDA, Java, SQL, JavaScript (Node.js, Next.js).

AI/ML: PyTorch, Keras, Core-ML, **Scikit-learn**, Pandas, NumPy, Computer Vision, Signal Processing, Model Optimization.

Cloud & Tools: AWS, Azure, Kubernetes, Docker, SLURM, HPC, Power BI, Git, GitHub, Linux, REST APIs.